

Sonal Singh

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PROFESSIONAL SUMMARY

Full-Stack Developer with 2+ years of experience in building Responsive UIs, scalable backend systems, data pipelines, cloud-native applications and LLM-powered workflows using Java, Spring Boot, JavaScript. Developed RESTful APIs, optimized SQL database (PostgreSQL, MySQL), and frontend developers create responsive designs. Committed to delivering high-performance code that drives measurable business growth. Collaborated with cross-functional teams and can translate complex business logic and requirements into clean and maintainable code, clear technical specifications, documentation as part of an Agile team.

WORK EXPERIENCE

Associate Developer

PetroIT Software Solutions

August 2024 – Present

Gurugram, India

- Architected and spearheaded the full-stack development of an enterprise platform serving 5,000+ users, utilizing Java (Spring Boot) and PostgreSQL to handle complex business logic and high-concurrency data workflows.
- Engineered a robust RBAC (Role-Based Access Control) framework using Spring Security; reduced security-related support tickets by 25% and ensured 100% compliance with enterprise audit standards.
- Optimized PostgreSQL database schemas and refactored SQL queries, resulting in a 40% reduction in query response times and improved overall system reliability.
- Developed a real-time push notification system using Web Push API, which increased user engagement by 35% and provided critical alerts for business-critical events.
- Managed cloud infrastructure on AWS (EC2, S3) and automated deployment workflows using Docker and GitHub Actions, reducing deployment downtime and streamlining CI/CD pipelines.
- Collaborated in an Agile environment to translate complex business requirements into technical specifications, providing production support and root-cause analysis (RCA) for mission-critical APIs.

Software Developer Intern

PetroIT Software Solutions

February 2024 – July 2024

Gurugram, India

- Integrated Apex-Charts for interactive data visualization (bar, line, pie, area), replacing static CSV exports and improving stakeholder decision-making
- Optimized 20+ SQL queries and backend data workflows using indexing strategies, resulting in a 40% decrease in retrieval.
- Built 20+ responsive, accessible UI components with UI/UX designers, ensuring pixel-perfect rendering across mobile, tablet, and web
- Maintained 95%+ code quality score in SonarQube through daily stand-ups, sprint planning, and bi-weekly code reviews

PROJECTS

TaskFlow AI — AI-Powered Task Management Platform

2024 – Present

Spring Boot, Next.js, TypeScript, PostgreSQL, Redis, Spring AI, OpenAI GPT-4o-mini, WebSocket, JWT

- Constructed full-stack AI task management platform with real-time multi-user collaboration using WebSocket (STOMP protocol)
- Secured JWT auth with 15-min access tokens, 7-day refresh token rotation, and token blacklisting via Redis for sub-ms validation
- Integrated Spring AI with GPT-4o-mini for intelligent task suggestions, auto-prioritization, and natural language categorization
- Designed Redis caching (5-min TTL) and rate limiting (100 req/min), reducing PostgreSQL read load by 60% under concurrent usage

Receipt Slayer — AI-Powered Expense Management App

2024 – Present

React Native, Node.js, PostgreSQL, Claude API, OpenAI, OCR

- Engineered a multi-model AI pipeline integrating Claude and OpenAI APIs to automate expense tracking via OCR and automated LLM categorization.
- Designed a RAG (Retrieval-Augmented Generation) workflow using vector databases to enable semantic search across processed financial documents.
- Containerized backend services using Docker and implemented CI/CD workflows for streamlined deployments and testing

EDUCATION

Bachelor of Engineering in Computer Science

Chandigarh University

August 2020 – July 2024

Mohali, India

TECHNICAL SKILLS

Languages: Java, Python, React, JavaScript, TypeScript, SQL, HTML5, CSS3, jQuery.

Frameworks: Spring Boot, Spring MVC, Spring Data JPA, Spring Security, Spring AI, Hibernate, Next.js, React, Bootstrap, Maven.

Databases: PostgreSQL, MySQL, MongoDB, Redis, Rabbit MQ, Firebase, Supabase.

Backend & Web: RESTful APIs, MVC Architecture, Microservices, JWT/Session Auth, WebSocket (STOMP), CRUD, JSON/API Integration, Data Ingestion, Debugging, Backend Automation.

AI/LLM: RAG, LLM Integration, MCP, Prompt Engineering, AI Chatbot Development, Vector Databases, OpenAI, Gemini CLI, Open-claw, Open source LLMs (Qwen, Mistral, Gemma)

Tools & Automation: Git, GitHub, GitHub Actions, Docker, CI/CD Pipelines, AWS EC2, S3, Cloud Deployments, IntelliJ IDEA, VS Code, Postman, n8n, Webhooks, Linux, Production Support.